

## Trainings ▾

## Select Service Tools

### Recommended Cummins® Service Tools

- No recommended Cummins® service tools

### Additional Service Items

- No additional service items required

## Preparatory Steps

### **⚠ WARNING ⚠**

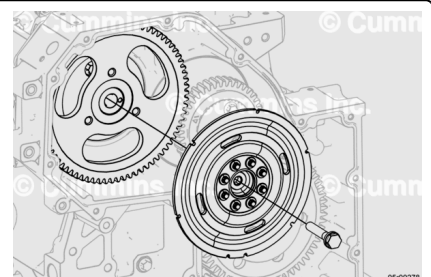
**Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.**

- Disconnect the batteries. See equipment manufacturer service information.
- Remove the upper gear cover. Refer to Procedure 001-079 in Section 1.  
(/qs3/pubsys2/xml/en/procedures/377/377-001-079.html)
- Remove the camshaft position sensor. Refer to Procedure 019-363 in Section 19.  
(/qs3/pubsys2/xml/en/procedures/377/377-019-363.html)

## Remove

### **⚠ CAUTION ⚠**

**The camshaft tone wheel can be disassembled from the camshaft gear. However, the camshaft tone wheel should not be disassembled from the mounting adapter. Damage to the tone wheel will result.**



**Note :** This procedure is for removing the camshaft tone wheel **only**. If the valve camshaft gear is to be removed, the following procedure **must** be used. Refer to Procedure 001-088 in Section 1.

(/qs3/pubsys2/xml/en/procedures/377/377-001-088.html)

Remove the valve camshaft gear mounting capscrew and camshaft tone wheel.

## Inspect

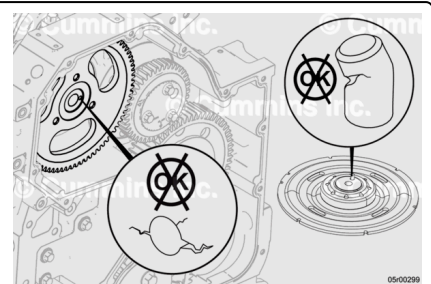
Inspect the machined surfaces for scratches or nicks.

Inspect the tone wheel alignment dowel for damage. If damaged, the dowel **must** be replaced.

Inspect the tone wheel alignment dowel hole for damage. If the dowel hole or the area surrounding the dowel hole has been damaged, the camshaft **must** be replaced.

Inspect the valve camshaft gear for excessive wear or damage.

If any teeth on the camshaft gear are missing, chipped, or pitted, the camshaft gear **must** be replaced.



## Install

Install the camshaft tone wheel assembly and valve camshaft gear mounting capscrew.

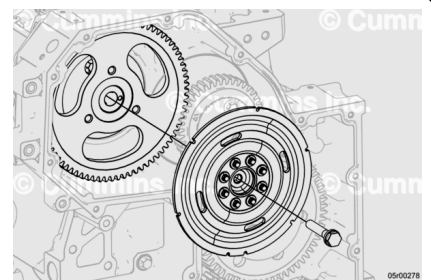
Tighten the capscrew.

### Torque Value:

1. 102 n•m [ 75 ft-lb ]
2. Plus 120 degrees

Check the valve gear to idler gear backlash. Refer to Procedure 001-055 in Section 1.

(/qs3/pubsys2/xml/en/procedures/377/377-001-055.html)



## Finishing Steps

### WARNING

**Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.**

- Install the camshaft position sensor. Refer to Procedure 019-363 in Section 19.  
(/qs3/pubsys2/xml/en/procedures/377/377-019-363.html)
- Install the upper gear cover. Refer to Procedure 001-079 in Section 1.  
(/qs3/pubsys2/xml/en/procedures/377/377-001-079.html)
- Connect the batteries. See equipment manufacturer service information.
- Operate the engine to normal operating temperature and check for leaks and proper operation.